

INDUSTRIAL PROGRAMMING LANGUAGE

MDA 3044

LAB 6

Name	KAUTHAR BINTI MOHD HISHAM
Matric No.	S76156
Program	BACHELOR OF SCIENCES (DATA ANALYTICS)
Group	K1

QUESTION 1

```java
package lab6;
import java.util.Scanner;

public class Q1 {

    public static long factorial(int n) {
            long result = 1;
            for (int i = 1; i <= n; i++) {
                result *= i;
            }
            return result;
        }

        public static long computeFact(int number) {
            return (number == 0 || number == 1) ? 1 : number * computeFact(number - 1);
        }

        public static void main(String[] args) {
            Scanner input = new Scanner(System.in);

            System.out.print("Enter a non-negative integer: ");
            int n = input.nextInt();

            System.out.println("Factorial using loop: " + factorial(n));
            System.out.println("Factorial using recursion: " + computeFact(n));
        }
}
```

```
Enter a non-negative integer: 5
Factorial using loop: 120
Factorial using recursion: 120
```

QUESTION 5

```java
package lab6;

public class Q5 {

        public static double sqrt(double n) {
            double lastGuess = 1.0;
            double nextGuess = (lastGuess + n / lastGuess) / 2.0;

            while (Math.abs(nextGuess - lastGuess) >= 0.0001) {
                lastGuess = nextGuess;
                nextGuess = (lastGuess + n / lastGuess) / 2.0;
            }

            return nextGuess;
        }

        public static void main(String[] args) {
            java.util.Scanner input = new java.util.Scanner(System.in);
            System.out.print("Enter a number to compute its square root: ");
            double number = input.nextDouble();
            System.out.printf("Approximated square root: %.2f", sqrt(number));
        }
}
```

```
Enter a number to compute its square root: 500
Approximated square root: 22.36
```

QUESTION 6

```java
package lab6;

import java.util.Scanner;

public class Q6 {

    public static double area(int n, double side) {
        return (n * side * side) / (4 * Math.tan(Math.PI / n));
    }

    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter number of sides: ");
        int n = input.nextInt();

        System.out.print("Enter the length of each side: ");
        double side = input.nextDouble();

        System.out.printf("Area of the polygon: %.2f\n", area(n, side));
    }
}
```

```
Enter number of sides: 7
Enter the length of each side: 20
Area of the polygon: 1453.56
```